

Online Research Seminar Syllabus

1. Overview

Title	AI in Finance		
Mode	Leading Instructor Sessions & Teaching Fellow Sessions		
Hours	4*1.5 hours lecture +2*2 workshops+1*2 hours final presentation session+ 6*1.5 hours teaching fellow sessions		
Prerequisites	High School Students	Required course/Knowledge	Calculus, some Probability, some Programming
		Recommended Materials for preparing for the course	
	College Students	Required course/Knowledge	Calculus, some Probability, some Programming
		Recommended Materials for preparing for the course	

2. Program Introduction and Objectives

Course Description	This course explores the dynamic intersection of financial technology (FinTech) and household credit markets, blending theoretical frameworks with hands-on data analysis. Students will gain a comprehensive understanding of how FinTech disrupts traditional lending practices, with a focus on credit risk modeling, machine learning applications, and ethical challenges in AI-driven finance. Through case studies, technical exercises, and a capstone project, participants will learn to critically evaluate lending innovations while developing practical skills in data-driven decision-making.
Software/Tools (if any)	Python, statistical libraries (e.g., scikit-learn), and visualization tools (Tableau/Matplotlib).

3. Program Schedule

Week		Leading Instructor Session	Teaching Fellow Session (lab/case study, etc.)	Reading Materials
1	Topic	Household credit overview	Q&A+Examples	Session 1 - https://www.dropbox.com/scl/fi/exmkw16ze0p0ml2grubh2/1-Has-the-US-Finance-Industry-Become-Less-Efficient.pdf?rlkey=2wz5e5lt82iidhcta9ywk8bg&dl=0
	Detail	We will start with a high-level context for what FinTech is and where it matters. We will proceed to look at lending markets within it, specifically, household credit. To that end, we look at the composition of the very large household credit market in the US and how fintech players innovate in it. Then, we start diving into credit default by developing a framework that allows us to map the business problems faced by lenders into quantifiable model performance measure.		
2	Topic	Building a baseline risk-based model	Q&A+Examples	Session 2 - Data and code: https://www.dropbox.com/scl/fo/uh6fwjzwvc0gtjbuny14d/AGaC0bGL2WiEt7XwVAzvgSY?rlkey=va19vhvffe992kat69y3m19hg&dl=0 - Chapter 4 of the book https://www.dropbox.com/scl/fi/ybw05xtxb14oons0pnqdx/3-Chapter-4-An-Introduction-to-Statistical-Learning.pdf?rlkey=cuandzm0j2chjwljnkktolpw6&dl=0 - Article about LC: https://www.crowdfundinsider.com/2024/12/233938-lendingclub-founding-ceo-and-current-upgrade-ceo-renaud-laplanche-comments-on-10-year-anniversary-of-iconic-fintech-ipo/
	Detail	We turn to look at the data we will use in this course, coming from LendingClub (NYSE: LC). To better understand the data, we will discuss the marketplace lending model that LC pioneered. We then explore the data and develop a baseline model using FICO scores.		
3	Topic	Machine learning in lending	Q&A+Examples	Session 3 - Chapter 8 of the book https://www.dropbox.com/scl/fi/ybw05xtxb14oons0pnqdx/3-Chapter-4-An-Introduction-to-Statistical-Learning.pdf?rlkey=cuandzm0j2chjwljnkktolpw6&dl=0
	Detail	In this session, we develop a machine learning model from the group up. Specifically, we start by establishing how decision trees are estimated, discuss their pros and cons, and proceed to build the Random Forest model and show its performance relative		

		to the baseline model in predicting defaults.		
4	Topic	Open questions at the intersection of AI and lending	Q&A+Examples	Session 4 - https://www.dropbox.com/scl/fi/81q445dygeki98rav5ag8/iv-1-1.pdf?rlkey=u1fbfnpeq53oskmfe3z8&dl=0 - https://www.dropbox.com/scl/fi/xrsl2scwh5vibjdkscyc2/wp24-09.pdf?rlkey=jlfvoeh35fxyz2ugr7c0lndej&dl=0
	Detail	In this session we dive into some of the most current questions at the intersection of AI and lending. Specifically, the debate about the benefits and costs introduced by AI, the open questions and challenges around discrimination and the frontier studies on explainable AI (“XAI”).		
5	Topic	Original analysis proposals - I	Research Workshop Group 1: Continued discussion of research workshop (remaining students/groups).	
	Detail	Sessions 5&6 will be dedicated to discussing group proposals and initial results. The objective is to develop a data-driven research question and conduct a basic empirical analysis using LendingClub loan-level data. Groups are expected to apply visualization, statistical methods and machine learning to extract insights related to credit risk, borrower behavior, or platform dynamics. Examples for questions can be: 1. Can alternative borrower features improve default prediction beyond FICO scores? Rather than trying to use all available variables, use a subset of variables and develop deeper understanding of their role. 2. Can augmented data help improve default predictions? For example, how important are local economic conditions where the borrower is located.		

		3. How do interest rates compare to actual default rates across different borrower segments? Focus on Insights, Not Complexity: Groups are encouraged to prioritize clear, interpretable results over complex modeling.		
6	Topic	Original analysis proposals - II	Research Workshop Group 2: Continued discussion of research workshop (remaining students/groups).	
	Detail	In this session, each group would have an opportunity to show its progress and receive feedback.		
7	Final Oral Presentation and Written Reporting			

4. Problem Sets/Written Assignments/Quizzes

None.

5. Final Oral and Written Project

Total Number of Assignments	___ 1 final project
Submission Deadline	Students will need to present the their final project orally in the final presentation.

Detailed requirements of the final project:

Develop a data-driven research question and conduct a basic empirical analysis using LendingClub loan-level data. Groups are expected to apply visualization, statistical methods and machine learning to extract insights related to credit risk, borrower behavior, or platform dynamics. Examples for questions can be:

- Can alternative borrower features improve default prediction beyond FICO scores? Rather than trying to use all available variables, use a subset of variables and develop deeper understanding of their role.
- Can augmented data help improve default predictions? For example, how important are local economic conditions where the borrower is located.
- How do interest rates compare to actual default rates across different borrower segments?

5.1 Final Oral Presentation

- Oral Project Requirements (e.g: if slides needed; Format; Criteria; Deadline):

Presentation Slides

1. Format: Use PowerPoint, Google Slides, or any other presentation tool.
2. Slides should be clear, visually appealing, and easy to read.
3. Include the following sections:
 - Title Slide: Project title, team members, and date.
 - Introduction: Problem statement and objectives.
 - Methodology: Techniques, tools, and models used.
 - Results: Key findings, visualizations, and outputs.
 - Conclusion: Summary of the project and future improvements.
 - Q&A: A slide inviting questions from the audience.
4. Length: 8-10 slides (8-10 minutes presentation).
5. Deadline: Slides must be submitted 2 days before the presentation date.

5.2 Will you require a written final report as well?

- Written Project Requirements (e.g: word count; style; criteria; Deadline):

Optional : DDL – one week after class 7

Format:

1. Use a standard document format (e.g., Word, PDF).
2. Include the following sections:
 - Title Page: Project title, team members, and submission date.
 - Abstract: A brief summary of the project (100-150 words).
 - Introduction: Problem statement, objectives, and relevance.
 - Methodology: Detailed explanation of techniques, tools, and models used.
 - Results: Analysis of findings, supported by tables, graphs, or screenshots.
 - Discussion: Interpretation of results, challenges faced, and lessons learned.
 - Conclusion: Summary of the project and potential future work.
 - References: List of sources (if any).
 - Word Count: 1,500-2,000 words.

6. Evaluation

Percentage basis of evaluation (must total 100%):

Class Participation: __30%__;

Final Project: ____70%____

7. Suggested Future Research Fields/Direction/Topics